
QUANTIZATION OF GENERATIVE ADVERSARIAL NETWORKS FOR EFFICIENT INFERENCE: A METHODOLOGICAL STUDY

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ABSTRACT

Generative adversarial networks (GANs) have an enormous potential impact on digital content creation, e.g., photo-realistic digital avatars, semantic content editing, and quality enhancement of speech and images. However, the performance of modern GANs comes together with massive amounts of computations performed during the inference and high energy consumption. That complicates, or even makes impossible, their deployment on edge devices. The problem can be reduced with *quantization*—a neural network compression technique that facilitates hardware-friendly inference by replacing floating-point computations with low-bit integer ones. While quantization is well established for discriminative models, the performance of modern quantization techniques in application to GANs remains unclear. GANs generate content of a more complex structure than discriminative models, and thus quantization of GANs is significantly more challenging. To tackle this problem, we perform an extensive experimental study of state-of-art quantization techniques on three diverse GAN architectures, namely StyleGAN, Self-Attention GAN, and CycleGAN. As a result, we discovered practical recipes that allowed us to successfully quantize these models for inference with 4/8-bit weights and 8-bit activations while preserving the quality of the original full-precision models.

Keywords: quantization, generative adversarial networks, compression

1 Introduction

Generative adversarial networks (GANs) are the tool of choice for a variety of computer vision tasks involving generative modeling [15, 32, 33]. However, the expressiveness of modern deep neural networks comes along with tremendous computational and memory resources spent during the inference phase, with GANs being a clear illustration of this tendency [2]. This circumstance significantly complicates the deployment of such models in real-world applications, especially on edge devices, where memory and latency are of the main concern.

Over the past few years, considerable research attention has been dedicated to the compression of deep neural networks. The most popular compression techniques include pruning [8, 20], efficient architecture design [4, 11], knowledge distillation [10, 3], and quantization [16, 29]. The last one is of particular research attention due to several reasons. First, it is “orthogonal” to the majority of other compression techniques, meaning that quantization can be applied